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Software Requirements Specification (SRS)

Project Title: Smart Truck Booking & Logistics Marketplace System

Prepared by: Particle14 Infotech Pvt. Ltd. Project Type: AI-Based Mobile & Web Application Version: 1.0 Date:

1. Introduction

1.1 Purpose

This document defines the requirements for a **Truck Booking & Logistics Mobile Application** that enables customers to book trucks and drivers/fleet owners to accept or bid on jobs.

1.2 Scope

The system will:

- Connect customers with truck drivers
- Provide real-time tracking
- Enable bidding and fixed pricing models
- Handle payments, KYC, and trip management

2. User Roles

2.1 Customer (Shipper)

- Book trucks
- Track shipments
- Make payments
- View booking history

2.2 Driver / Fleet Owner

- Accept or bid on jobs

- Manage vehicles
- Track earnings
- Upload documents

2.3 Admin (Web Panel)

- Manage users & drivers
- Verify KYC
- Control pricing & commissions
- Handle disputes

3. System Features

3.1 Customer Application

3.1.1 Splash & Onboarding

- 3-step animated onboarding
- App introduction screens

3.1.2 Authentication

- OTP-based login
- Auto resend countdown
- Secure session management

3.1.3 Home Dashboard

- Active shipments display
- Truck categories
- Recent bookings

3.1.4 Book a Truck

- Pickup & drop location
- Multi-stop option
- Schedule booking (instant/future)

3.1.5 Truck Selection

- 6 truck categories
- Pricing breakdown per type

3.1.6 Load Details

- Goods type selection
- Weight input
- Insurance toggle

3.1.7 Fare Estimation

- Full fare breakdown
- Fixed pricing mode
- Bidding mode

3.1.8 Driver Matching

- Show nearby drivers (min 3)
- ETA, rating, and price

3.1.9 Live Tracking

- Real-time GPS tracking
- Animated route map
- Progress bar
- Driver contact option

3.1.10 Payment System

- Wallet integration
- 4 payment methods
- Promo code support

3.1.11 Booking History

- Filter by status
- Re-book option
- Tracking history

3.1.12 Profile & Settings

- Saved addresses
- Customer support
- Account management

3.2 Driver / Fleet Owner Application

3.2.1 Role Selection

- Driver or Fleet Owner

3.2.2 Authentication

- OTP login
- Verification workflow

3.2.3 Dashboard

- Online/offline toggle

- Earnings overview
- Job alerts

3.2.4 Job Requests

- Accept / Reject jobs
- Place bids on trips

3.2.5 Active Trip

- Navigation map
- Contact shipper
- Trip completion

3.2.6 Earnings & Wallet

- Earnings dashboard
- Transaction history
- Withdraw funds

3.2.7 Trip History

- Completed trips
- Ratings & reviews

3.2.8 KYC & Documents

- Upload 8 document types
- Verification status tracking

3.2.9 Vehicle Management

- Add/edit truck details
- Working preferences

3.2.10 Ratings & Reviews

- 5-star rating system
- Driver responses

3.2.11 Notifications

- Push notifications
- Job alerts

3.2.12 Bidding Module

- Compete with other drivers
- Real-time bid updates

3.3 Admin Panel (Web)

3.3.1 Dashboard

- Total users, drivers, bookings
- Revenue analytics

3.3.2 User Management

- View/manage customers & drivers
- Block/unblock accounts

3.3.3 KYC Verification

- Approve/reject documents
- Manual verification system

3.3.4 Booking Management

- Monitor all bookings
- Cancel/modify trips

3.3.5 Pricing Control

- Base fare settings
- Distance pricing
- Commission control

3.3.6 Payment Management

- Transaction tracking
- Refund handling

3.3.7 Dispute Management

- Raise & resolve disputes

3.3.8 Notification System

- Send push notifications

4. Non-Functional Requirements

4.1 Performance

- Real-time updates within 2–3 seconds

4.2 Security

- OTP authentication
- Encrypted transactions
- Secure APIs

4.3 Scalability

- Support 10,000+ concurrent users

4.4 Availability

- 99.5% uptime

5.Third-Party Integrations

- Google Maps API (GPS tracking)
- Payment Gateway (Razorpay / Stripe)
- Firebase (Auth, Notifications)
- SMS Gateway (OTP)

6. Database Overview

Entities:

- Users
- Drivers
- Vehicles
- Bookings
- Payments
- Bids
- Reviews
- Documents

7. System Workflow

1. User logs in
2. Books a truck
3. Selects pricing mode (Fixed/Bidding)
4. Driver accepts or bids
5. Trip starts
6. Live tracking
7. Trip completes
8. Payment processed
9. Rating & review

8. Deliverables

Mobile Application
Web Application (if required)
Admin Dashboard
Backend APIs
Deployment Setup
Launch Support

Prepared By:

Neeraj Kumar, CEO

A handwritten signature in black ink, appearing to read 'Neeraj', is written over a long, horizontal, slightly wavy line that serves as a baseline for the signature.

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